Practice questions

Week 1 readings and work: De Angelis, Bryan, Wilson et al., and the shelter data

Rohan Alexander

16 September, 2026

1. Bryan (2018) says Git works best with one kind of file and badly with another. Name both kinds, and give one example of a file from your assignment repo that you would think twice about committing.

2. Wilson et al. (2017) organise their recommendations under six headings. Name three of them.

3. Wilson et al. give a rule for raw data. What is it, and where in the project layout they recommend does raw data live relative to the files your scripts generate?

4. De Angelis (2026) says a GitHub Actions workflow file has five core components. Name three of them, and say in one line what one of the three does.

5. In a scheduled GitHub Actions workflow, `cron: '0 7 * * *'` runs at 7:00 in which time zone? And which repository setting has to be changed before the workflow can commit the data it collects back to the repo?

6. Where does De Angelis store the Bluesky username and password, and how does the YAML file refer to them?

7. The Week 1 demonstration used the daily shelter overnight occupancy data from Open Data Toronto.

   a. What does one row of the raw file represent?
   b. The data distinguish two capacity types. Name them, and say why summing `OCCUPIED_BEDS` does not give the number of people in the system.
   c. How many people were in Toronto's shelter system last night?

8. Your Actions workflow installs `uv` and runs `uv sync` before running the script, even though the script already runs on your laptop. Why?

9. Suggest one question that would assess student learning of any of the readings.

References

Bryan, Jennifer. 2018. "Excuse Me, Do You Have a Moment to Talk about Version Control?" *The American Statistician* 72 (1): 20–27. https://doi.org/10.1080/00031305.2017.1399928.

De Angelis, Inessa. 2026. *Using GitHub Actions for Computational Communication Research.* SocArXiv. https://doi.org/10.31235/osf.io/uqf6n_v1.

Wilson, Greg, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. 2017. "Good Enough Practices in Scientific Computing." *PLOS Computational Biology* 13 (6): e1005510. https://doi.org/10.1371/journal.pcbi.1005510.